

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Repeat measurements OR measure more than 1 wavelength Don't accept reference to the ruler or more accurate ripple tank			1	1		1
		(ii)		Wave speed is [directly] proportional to wavelength (1) As wave speed 2 times faster in deep water or 2 times slower in shallow water (1) so teacher is incorrect Alternative 1: Candidates may make up a freq and use it on both sides e.g. 4 Hz In shallow water $4 \times 5 = 20$ [mm/s] and in deep water $4 \times 10 = 40$ [mm/s] (1) As wave speed 2 times faster in deep water or 2 times slower in shallow water (1) so teacher is incorrect Alternative 2: Wave in deep water travels double the distance (1) in the same time compared to shallow water (1) so teacher is incorrect N.B. To award both marks conclusion must be present. Award 1 mark only Waves in shallow water travel slower than waves in deep water so teacher claim is incorrect			2	2	2	2
	(b)	(i)		<u>Total internal reflection</u> accept TIR	1			1		
		(ii)		Light [ray directed] from high density to low density (1) <u>Angle</u> [of incidence] > critical angle or <u>angle</u> [of incidence] > 42° (1)	2			2		
		(iii)		CT scans are ionising or X-rays are ionising (1) and [X-rays] mutate cells / cause cancer (1) OR [Endoscope uses] light that is not ionising (1) and [light] doesn't mutate cells / doesn't cause cancer (1)	2			2		
				Question 2 total	5	0	3	8	2	3